

Clinical Reference Guide: Small Intestine Cancer

Extracted Focus & Question-Answer Summary from Cancer.gov PDQ

DOCUMENT OVERVIEW

Focus Condition:	Small Intestine Cancer
Semantic Group:	Disorders (T191)
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Question 1: What is (are) Small Intestine Cancer ?

Small intestine cancer is a rare disease in which malignant (cancer) cells form in the tissues of the small intestine. The small intestine is a long folded tube connecting the stomach to the large intestine that removes and processes vitamins, minerals, carbohydrates, fats, proteins, and water from foods and helps pass waste material out of the body.

Primary Tumor Types: There are five types of cancer found in the small intestine: adenocarcinoma, sarcoma, carcinoid tumors, gastrointestinal stromal tumor, and lymphoma. This clinical summary highlights two major classifications:

- **Adenocarcinoma:** The most common type of small intestine cancer. It starts in glandular cells in the lining of the small intestine, most frequently occurring in the section near the stomach where it may grow and cause an intestinal obstruction.
- **Leiomyosarcoma:** A type of sarcoma that originates in the smooth muscle cells of the small intestine. Most of these tumors develop in the portion of the small intestine located near the large intestine.

Question 2: Who is at risk for Small Intestine Cancer? ?

Diet and personal health history play a significant role in influencing an individual's susceptibility to developing small intestine cancer. While having a risk factor does not guarantee malignancy, established risk factors include:

- Eating a high-fat diet.
- Having Crohn disease.
- Having celiac disease.
- Having familial adenomatous polyposis (FAP).

Question 3: What are the symptoms of Small Intestine Cancer ?

Signs and symptoms of small intestine cancer can resemble benign gastrointestinal conditions. Patients should consult a physician if they experience any of the following manifestations:

- Pain or cramps in the middle of the abdomen.
- Unexplained weight loss with no known reason.
- A palpable lump or mass in the abdomen.
- Blood present in the stool.

Question 4: How to diagnose Small Intestine Cancer ?

Tests and procedures that examine the small intestine are used simultaneously to detect, diagnose, and stage small intestine cancer, helping doctors identify the tumor type and determine surgical resectability:

- **Physical Exam and Medical History:** Assessment of overall health, past illnesses, and clinical inspection for abdominal lumps or anomalies.
- **Laboratory Analyses:** Includes *Blood Chemistry Studies* and *Liver Function Tests* (measuring substances released by organs; elevated liver enzymes can signal liver disease or metastasis).
- **Endoscopic Procedures:** Utilized to visually inspect and biopsy intestinal mucosa:
 - *Upper Endoscopy:* Examines the esophagus, stomach, and duodenum (first part of the small intestine).
 - *Capsule Endoscopy:* The patient swallows a pill-sized wireless camera capsule that transmits detailed images of the entire small intestine to a wearable recorder.
 - *Double Balloon Endoscopy (Enteroscopy):* Uses a specialized double-tube instrument with inflatable balloons to navigate and examine deep sections of the small intestine, allowing tissue sample removal when capsule endoscopy results are abnormal.
- **Surgical Exploration (Laparotomy):** A surgical incision into the abdomen to inspect internal organs and remove lymph nodes or tissue samples for histological analysis.
- **Biopsy:** The definitive diagnostic standard, involving microscopic evaluation of cellular tissue obtained during endoscopy or laparotomy.
- **Radiographic & Imaging Modalities:** Includes an *Upper GI Series with Small Bowel Follow-Through* (serial x-rays taken as liquid barium travels through the GI tract), computed tomography (CT/CAT scans), and magnetic resonance imaging (MRI).

Question 5: What is the outlook for Small Intestine Cancer ?

The prognosis (probability of recovery) and the selection of appropriate therapeutic pathways depend on several crucial clinical factors:

- **Tumor Type:** The histological classification of the small intestine cancer.

- **Invasion Depth:** Whether the cancer is confined to the inner mucosal lining or has invaded into and beyond the small intestine wall.
- **Metastatic Spread:** Whether cancer cells have disseminated to lymph nodes, the liver, or the peritoneum (abdominal lining).
- **Surgical Resectability:** Whether the tumor can be completely excised by surgery.
- **Disease Status:** Whether the malignancy is a newly diagnosed primary lesion or a recurrence.

Question 6: What are the stages of Small Intestine Cancer ?

Staging determines the extent of cancer dissemination through tissue invasion, the lymphatic system, or the bloodstream (metastasis). In clinical management, small intestine cancer is categorized primarily by surgical resectability rather than a numerical stage alone:

- **Resectability Grouping:** Treatment pathways depend heavily on whether the tumor can be completely removed by surgery and whether it represents a localized primary tumor or metastatic disease.
- **Metastatic Disease:** When cancer breaks away from the primary tumor in the small intestine and spreads to distant organs (e.g., the liver), it is categorized as metastatic small intestine cancer, not a primary cancer of the receiving organ.

Question 7: What are the treatments for Small Intestine Cancer ?

Management of small intestine cancer relies on standard therapeutic modalities and active clinical research trials:

- **Surgery:** The most common primary treatment:
 - *Resection & Anastomosis:* Surgical removal of the tumor-containing bowel segment, adjacent tissues, and regional lymph nodes, followed by joining the cut intestinal ends (anastomosis).
 - *Bypass Surgery:* Performed when a tumor blocking the intestine cannot be removed, routing food around the obstruction.
 - *Adjuvant Therapy:* Post-operative radiation given to eliminate microscopic residual disease and reduce recurrence risk.
- **Radiation Therapy:** Deploys external beam radiation (using machines outside the body) or internal radiation to destroy cancer cells. Palliative radiation is often used for unresectable tumors to relieve symptoms.
- **Chemotherapy:** Systemic or regional administration of anti-cancer drugs to halt cellular division. It is often utilized as palliative therapy or in combination regimens.
- **Clinical Trials & Emerging Therapies:** Active research is evaluating new anti-cancer drugs, *Biologic Therapy* (using the patient's immune system to fight cancer), and *Radiation Therapy with Radiosensitizers* (administering drugs that make tumor cells more sensitive to radiation).

- **Recurrent Disease Management:** Locally recurrent tumors may be treated with repeat surgery, palliative chemo/radiation, or clinical trials. Metastatic recurrences are generally managed via clinical trials evaluating novel drugs or immunotherapy.
- **Follow-Up Care:** Ongoing medical surveillance with regular blood work and imaging is mandatory to monitor treatment efficacy and catch potential recurrences early.

This document was generated for educational and clinical reference purposes based on official patient treatment guidelines (PDQ) from the National Cancer Institute.